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BUMPASS FIRE & RESCUE
PASSWORD CRYPTOGRAPHIC PROTECTION - CONFIGURATION AUDIT
Evidence for CMMC Control 3.5.10

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Report Date: July 25, 2026
Prepared By: Mike Sadler, IT Administrator
Domain: bfr.local

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AD PASSWORD POLICY - REVERSIBLE ENCRYPTION

Get-ADDefaultDomainPasswordPolicy output (relevant fields):
ReversibleEncryptionEnabled : False

Result: Passwords cannot be recovered from AD database.
COMPLIANT.

NTDS.DIT PROTECTION

Location: C:\Windows\NTDS\ntds.dit (on DC01)
Protection: Encrypted by DPAPI (Data Protection API)
Access: SYSTEM account only (ACL restricted)
VSS shadow copy access: Blocked for non-administrators
DSRM password rotation: Last changed June 1, 2026

AZURE AD PASSWORD HASH SYNC

Password Hash Sync: ENABLED (via Azure AD Connect)
Hash format: Double-hashed (MD4 + HMAC-SHA256)
Microsoft does not store original hash
Transmission: Encrypted HTTPS to Azure AD
Plaintext transmitted: NEVER

LDAP CONFIGURATION

LDAP (port 389) simple bind: DISABLED
LDAPS (port 636): ENABLED and required
GPO: BFR-Secure-LDAP
Network security: LDAP client signing requirements: Require signing
Domain controller: LDAP server signing requirements: Require signing

AZURE KEY VAULT - SERVICE ACCOUNT CREDENTIALS

Vault: bfr-keyvault
Service account passwords stored: 3 secrets
Plaintext password files: NONE found in quarterly scan

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Certified: Mike Sadler - July 25, 2026

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